The Contestable Frontier

Three Lenses on Why GenAI Markets Resist Concentration — and the Catches

Audit synthesis

Carugati (2023), *Bruegel WP 14/2023* | Jagadeesan, Jordan & Steinhardt (2025), *PNAS* 122(42):e2510004122
Zhang et al. (2024), *Working Paper* (Mihuashi / NovelAI leak)

$$-64\% \quad | \quad +121\% \quad | \quad +56\%$$

prices | volume | revenue, when AI flooded a creative-goods market.

Incumbent human artists captured **97.4%** of post-leak orders.

The concentration narrative misreads the frontier. Markets contest before they tip.

Zhang et al. (2024); NovelAI Stable Diffusion leak, Mihuashi platform, ~197,000 orders, 5 months post-shock.

Three lenses on whether GenAI markets will concentrate — and three different "no, but"s

Lens	Verdict on concentration	Catch
Policy	Today's FM market *competitive* (Carugati 2023)	5 future risks: leverage, license refusal, scraping, lock-in, collusion
Theory	Asymmetric safety lowers entry (Jagadeesan 2025)	Race-to-the-bottom on safety
Empirics	Output market *expanded* (Zhang 2024)	Prices crushed; gains low-end only

The frontier is contestable now. Whether it stays so depends on what gets regulated and how.

Lens 1 — Policy: today's market is competitive

but the future tail is heavy

The FM value chain has three inputs — and all three are currently competitive

Input	Status (mid-2023)	Why
Language models	multiple closed + open	Anthropic, Meta, Mistral, BAAI, OpenAI, Google
Data	diverse open + proprietary	Common Crawl, GitHub, scraped + licensed
Compute	vigorous despite top-3	AWS, Azure, GCP compete

Entry barriers "low or surmountable" — small models (SLMs) and Low-Rank Adaptation (LoRA, Hu et al. 2021) collapse training cost.

Carugati (2023), Bruegel WP 14/2023, Section 2; pre-DeepSeek.

Five tipping mechanisms could lock in incumbents tomorrow

Risk	Mechanism
Leverage	Dominant firm extends LM/cloud control into adjacent products
License refusal	Closed-LM holders deny competing application developers
Data lock-out	Refusal to grant data access; scraping wars
Switching costs	User lock-in via memory, context, integrated workflow
Algorithmic collusion	LMs converge on tacit pricing or content rules

Carugati's policy ask: protect open- vs. closed-source competition; do *not* regulate open-source so heavily that closed-source incumbents win by default.

Carugati (2023), Section 3 (competition issues), Section 5 (policy recommendations).

Lens 2 — Theory: asymmetric safety *helps* entrants

the very thing that protects users opens the door

Two firms, two objectives — entrant exploits a weaker safety threshold

Object	Definition
$\alpha \in [0.5, 1]$	Share of data labeled with performance (vs. safety) objective
$\rho \in [0, 1)$	Correlation between β_1 (perf) and β_2 (safety) regressors
τ_I, τ_E	Safety constraints; $\tau_E > \tau_I$ (regulatory asymmetry)
$\nu = 0.34$	Single-objective scaling exponent, calibrated to Chinchilla
N_E^*	Min. entrant data to match incumbent at τ_E

Theorem 1. Entrant can enter with *finite* data even at $N_I = \infty$, if τ_I binds and $\rho < 1$.

Jagadeesan, Jordan & Steinhardt (2025), *PNAS*; pure theory, no data.

Multi-objective scaling exponent collapses as data grows: $0.34 \to 0.25 \to 0$

Regime	Range of N	ν^* (Thm 2)	Calibrated
1	$N \leq (1-\alpha)^{-1/\nu}(1-\rho)^{-1/\nu}$	ν	0.34
2	intermediate	$\nu/(\nu+1)$	0.25
3	$N \geq (1-\alpha)^{-(2+\nu)/\nu}(1-\rho)^{-1/\nu}$	$\mathbf{0}$	0

Theorem 4 corollary. $N_E^* = \Theta(N_I^c)$, where $c = 1 \to 0.75 \to 0$ as $N_I \to \infty$. Past a threshold, more incumbent data buys nothing the entrant can't match cheaply.

Jagadeesan et al. (2025), Theorems 2, 4; calibration to Hoffmann et al. (2022) Chinchilla scaling.

The mechanism that lowers barriers also tempts a race-to-the-bottom on safety

Theorem 5 finding	Implication
$N_E^* = \Theta(D^{-c})$, $D = G_I - G_E$	As safety thresholds converge, entry threshold *explodes*
c rises from $1/\nu$ $(=2.94)$ to $(\nu'+1)/\nu'$ as $D\downarrow$	Entrant must dataset-scale ever harder

Policy corner. Stricter incumbent-only safety regulation (DSA's VLOP $\geq$ 45M users; FTC investigations) *lowers* entry barriers — but only if entrants' weaker safety is socially tolerable. Otherwise the gap closes and entry collapses.

Jagadeesan et al. (2025), Theorem 5; Section "Implications," p. 9.

Lens 3 — Empirics: a downstream market *expanded*

but the price compression was severe

When AI flooded a creative-goods market, prices fell 64% but revenue rose 56%

Outcome (DiD, TWFE)	Estimate	Effect size
Average price	-646.5 yuan	-64%
Weekly volume	$+140.3$ orders	$+121\%$
Weekly revenue	$+56{,}674$ yuan	$+56\%$

Treatment: tachie (anime character art). Control: wallpaper. Shock: NovelAI Stable Diffusion leak, 4chan, October 6, 2022. All coefficients significant at 1%.

Zhang et al. (2024), Table 2; Mihuashi platform, Jan 2022–Feb 2023; $N = 126$ category-weeks.

Incumbent human artists captured 97.4% of post-leak orders — they adapted, they were not displaced

Margin	Incumbents	New entrants
Post-leak orders	**97.4%**	2.6%
Post-leak revenue	**97.96%**	2.04%
Log volume, personal commissions	$+2.95$	—
Log volume, commercial commissions	$+0.39$	—

Expansion happened low-end: new personal buyers, not displaced commercial demand. Incumbents cut prices and raised volume — adaptation, not wipeout.

Zhang et al. (2024), Tables 3–4.

Devil's advocate: aren't all three findings just early-stage snapshots?

The strongest objection

Carugati (2023) is pre-DeepSeek; Jagadeesan (2025) has *no data*; Zhang (2024) is 5 months, one platform, $N = 126$. The frontier might be contestable today and concentrated tomorrow.

Reply. The synthesis is not that concentration won't happen; it is that *whether* it happens depends on the policy choices each lens names:

- Carugati: regulators must protect open-source from over-restriction.
- Jagadeesan: $\tau_E > \tau_I$ is the lever; Theorem 5 shows converging τs collapse entry.
- Zhang: low-end expansion is real, but supplier surplus depends on platform fees.

The frontier is contestable *conditional on* these levers. Pull them wrong, the market tips.

The frontier is contestable today — the policy levers that decide tomorrow are explicit

Lens	Today's signal	Lever that decides tomorrow
Policy	3 inputs competitive	Open-source carve-out from regulation
Theory	$\tau_E > \tau_I$ admits finite-N entry	Whether τ asymmetry persists
Empirics	Prices -64%, volume $+121\%$	Platform fee structure on low-end orders

The same three levers work either way. The market does not concentrate by physics; it concentrates by choice.

Concentration is not a forecast.

It is a policy choice the frontier has not yet made.